Radian Measure and Arc Length

ANSWER KEY

Section 1: Degrees and Radians Conversion

1 Convert 150° to exact radians in simplest form.

a. $\frac{5\pi}{6}$ ✓

b. $\frac{2\pi}{3}$

c. $\frac{3\pi}{4}$

d. $\frac{7\pi}{6}$

$150^\circ \times \frac{\pi}{180^\circ} = \frac{150\pi}{180} = \frac{5\pi}{6}$.

2 Convert $\frac{7\pi}{4}$ radians to degrees.

a. 315° ✓

b. 225°

c. 300°

d. 330°

$\frac{7\pi}{4} \times \frac{180^\circ}{\pi} = 7 \times 45^\circ = 315^\circ$.

3 Find a positive angle coterminal with $-\frac{2\pi}{3}$ in the interval $[0, 2\pi)$.

a. $\frac{4\pi}{3}$ ✓

b. $\frac{\pi}{3}$

c. $\frac{5\pi}{3}$

d. $\frac{2\pi}{3}$

$-\frac{2\pi}{3} + 2\pi = -\frac{2\pi}{3} + \frac{6\pi}{3} = \frac{4\pi}{3}$.

Section 2: Arc Length

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4 State the formula for arc length s subtended by a central angle θ (in radians) in a circle of radius r.

a. $s = r\theta$ ✓

b. $s = \frac{1}{2}r^2\theta$

c. $s = 2\pi r\theta$

d. $s = \frac{r}{\theta}$

By the definition of radian measure, the arc length subtended by central angle θ is $s = r\theta$.

5 Find the arc length intercepted by a central angle of $\frac{2\pi}{3}$ in a circle of radius 9 cm.

a. 6π cm ✓

b. 3π cm

c. 18π cm

d. 12π cm

$s = r\theta = 9 \times \frac{2\pi}{3} = 6\pi$ cm.

6 An arc of length 10 m subtends a central angle of 2 radians. What is the radius of the circle?

a. 5 m ✓

b. 20 m

c. 10 m

d. 2.5 m

$s = r\theta \Rightarrow 10 = r(2) \Rightarrow r = 5$ m.

Section 3: Sector Area

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7 State the formula for the area A of a sector of a circle with radius r and central angle θ (in radians).

a. $A = \frac{1}{2}r^2\theta$ ✓

b. $A = r^2\theta$

c. $A = \frac{1}{2}r\theta$

d. $A = \pi r^2\theta$

The area of a sector proportional to the circle is $\frac{\theta}{2\pi}(\pi r^2) = \frac{1}{2}r^2\theta$.

8 Find the area of a sector with radius 6 in and central angle $\frac{\pi}{4}$.

a. $\frac{9\pi}{2}$ sq in ✓

b. 9π sq in

c. $\frac{9\pi}{4}$ sq in

d. 18π sq in

$A = \frac{1}{2}r^2\theta = \frac{1}{2}(36)(\frac{\pi}{4}) = 18(\frac{\pi}{4}) = \frac{9\pi}{2}$.

9 If a sector of radius 4 cm has an area of 8π cm^2, find its central angle θ.

a. π radians ✓

b. 2π radians

c. $\pi/2$ radians

d. 4π radians

$A = \frac{1}{2}r^2\theta \Rightarrow 8\pi = \frac{1}{2}(16)\theta = 8\theta \Rightarrow \theta = \pi$ radians.

Section 4: Linear and Angular Speed

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10 What is the relationship between linear speed v, radius r, and angular speed ω (in rad/time)?

a. $v = r\omega$ ✓

b. $v = \frac{\omega}{r}$

c. $v = r^2\omega$

d. $v = 2\pi r\omega$

Since arc length is $s = r\theta$, dividing by time t gives $\frac{s}{t} = r\frac{\theta}{t} \Rightarrow v = r\omega$.

11 A wheel of radius 2 ft rotates at 60 rpm. Find its angular speed in radians per second.

a. 2π rad/s ✓

b. 4π rad/s

c. π rad/s

d. 120π rad/s

$60 \text{ rev/min} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = 2\pi \text{ rad/s}$.

12 In the previous question ($r = 2$ ft, $\omega = 2\pi$ rad/s), what is the linear speed of a point on the rim?

a. 4π ft/s ✓

b. 2π ft/s

c. 8π ft/s

d. 120 ft/s

$v = r\omega = 2(2\pi) = 4\pi$ ft/s.

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